

Chapter 23

The Fast Fourier Transform and Applications

*Once the [FFT] method was established
it became clear that it had a long and interesting prehistory
going back as far as Gauss.
But until the advent of computing machines
it was a solution looking for a problem.*

— T. W. KÖRNER, *Fourier Analysis* (1988)

Life as we know it would be very different without the FFT.

— CHARLES F. VAN LOAN, *Computation/
Frameworks for the Fast Fourier Transform* (1992)

23.1. The Fast Fourier Transform

The matrix–vector product $y = F_n x$, where

$$F_n = \left(\exp(-2\pi i(r-1)(s-1)/n) \right)_{r,s=1}^n,$$

is the key computation in the numerical evaluation of Fourier transforms. If the product is formed in the obvious way then $O(n^2)$ operations are required. The fast Fourier transform (FFT) is a way to compute y in just $O(n \log n)$ operations. This represents a dramatic reduction in complexity.

The FFT is best understood (at least by a numerical analyst!) by interpreting it as the application of a clever factorization of the discrete Fourier transform (DFT) matrix F_n .

Theorem 23.1 (Cooley–Tukey radix 2 factorization). *If $n = 2^l$ then the DFT matrix F_n may be factorized as*

$$F_n = A_l \dots A_1 P_n, \quad (23.1)$$

where P_n is a permutation matrix and

$$\begin{aligned} A_k &= I_{2^{l-k}} \otimes B_{2^k}, \\ B_{2^k} &= \begin{bmatrix} I_r & \Omega_r \\ I_r & -\Omega_r \end{bmatrix}, \quad r = 2^{k-1}, \\ \Omega_r &= \text{diag}(1, \omega_k, \dots, \omega_k^{r-1}), \quad \omega_k = \exp(-2\pi i/2^k). \end{aligned}$$

Proof. See Van Loan [1044, 1992, Thm. 1.3.3]. $\square$

The theorem shows that we can write $y = F_n x$ as

$$y = A_l \dots A_1 P_n x.$$

which is formed as a sequence of matrix–vector products. It is the sparsity of the A_k (two nonzeros per row) that yields the $O(n \log n)$ operation count.

We will not consider the implementation of the FFT, and therefore we do not need to define the “bit reversing” permutation matrix P_n in (23.1). However, the way in which the weights ω_k^j are computed does affect the accuracy. We will assume that computed weights $\hat{\omega}_k^j$ are used that satisfy, for all j and k ,

$$\hat{\omega}_k^j = \omega_k^j + \epsilon_{kj}, \quad |\epsilon_{kj}| \leq \mu. \quad (23.2)$$

Among the many methods for computing the weights are ones for which we can take $\mu = cu$, $\mu = cu \log j$, and $\mu = cuj$, where c is a constant that depends on the method; see Van Loan [1044, 1992, §1.4].

We are now ready to prove an error bound.

Theorem 23.2. Let $\widehat{y} = fl(F_n x)$, where $n = 2^t$, be computed using the Cooley-Tukey radix 2 FFT, and assume that (23.2) holds. Then

$$\frac{\|y - \widehat{y}\|_2}{\|y\|_2} \leq n^{1/2} \frac{t\eta}{1 - t\eta}, \quad \eta := \mu + \gamma_4(1 + \mu).$$

Proof. Denote by $\widehat{A}_k$ the matrix defined in terms of the computed $\widehat{\omega}_k^j$. Then

$$\begin{aligned} \widehat{y} &= fl(\widehat{A}_t \dots \widehat{A}_1 P_n x), \\ &= (\widehat{A}_t + \Delta \widehat{A}_t) \dots (\widehat{A}_1 + \Delta \widehat{A}_1) P_n x, \quad |\Delta \widehat{A}_k| \leq \gamma_4 |\widehat{A}_k|, \end{aligned}$$

using the fact that each A_k has only two nonzeros per row, and recalling that we are using complex arithmetic. In view of (23.2),

$$\widehat{A}_k = A_k + \Delta A_k, \quad |\Delta A_k| \leq \mu |A_k|.$$

Hence, overall,

$$\begin{aligned} \widehat{y} &= (A_t + E_t) \dots (A_1 + E_1) P_n x, \\ |E_k| &\leq (\mu + \gamma_4(1 + \mu)) |A_k| = \eta |A_k|. \end{aligned}$$

Invoking Lemma 3.7 we find that

$$\begin{aligned} |y - \widehat{y}| &\leq [(1 + \eta)^t - 1] |A_t| \dots |A_1| P_n |x| \\ &\leq \frac{t\eta}{1 - t\eta} |A_t| \dots |A_1| P_n |x|, \end{aligned}$$

using Lemma 3.1 for the second inequality. Now $\| |A_k| \|_2 = \| |B_{2^k}| \|_2 = \left\| \begin{bmatrix} I & I \\ I & I \end{bmatrix} \right\|_2 = 2$. Hence

$$\|y - \widehat{y}\|_2 \leq \frac{t\eta}{1 - t\eta} 2^t \|x\|_2. \quad (23.3)$$

Finally, because F_n is $\sqrt{n}$ times a unitary matrix ($F_n^* F_n = nI$), $\|x\|_2 = n^{-1/2} \|y\|_2$. $\square$

Theorem 23.2 says that the Cooley–Tukey radix 2 FFT yields a tiny forward error, provided that the weights are computed stably. It follows immediately that the computation is backward stable, since $\widehat{y} = y + \Delta y = F_n x + \Delta y$ implies $\widehat{y} = F_n (x + \Delta x)$ with $\|\Delta x\|_2 / \|x\|_2 = \|\Delta y\|_2 / \|y\|_2$. If we form $y = F_n x$ by conventional multiplication using the exact F_n , then (Problem 3.7) $|y - \widehat{y}| \leq \gamma_{n+2} |F_n| |x|$, so $\|y - \widehat{y}\|_2 \leq n^{1/2} \gamma_{n+2} \|y\|_2$. Hence when μ is of order u , the FFT has an error bound smaller than that for conventional multiplication by the same factor as the reduction in complexity of the method. To sum up, the FFT is perfectly stable.

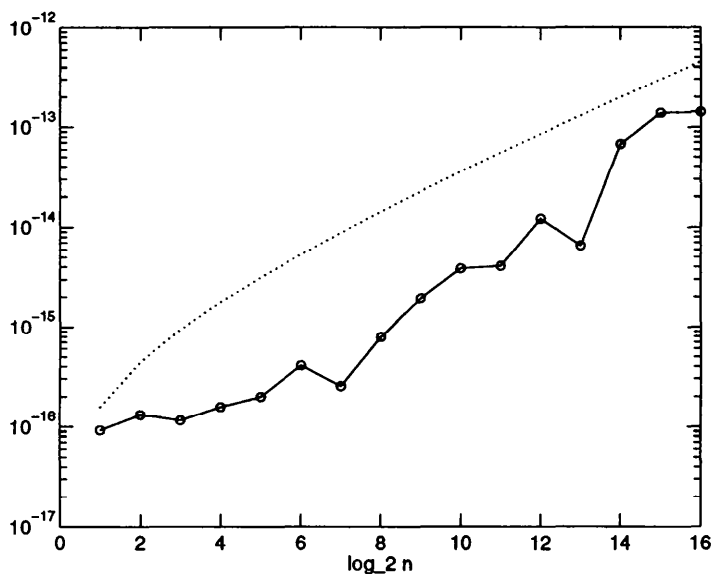


Figure 23.1. Error in FFT followed by inverse FFT (“o”). Dotted line is error bound.

The inverse transform $x = F_n^{-1}y = n^{-1}F_n^*y$ can again be computed in $O(n \log n)$ operations using the Cooley–Tukey radix 2 factorization, and $\|x - \hat{x}\|_2/\|x\|_2$ satisfies the same bound as in Theorem 23.2. (Strictly, we should replace t by $t + 1$ in the bound to account for the rounding error in dividing by n)

For other variations of the FFT, based on different radices or different factorization of F_n , results analogous to Theorem 23.2 hold.

A simple way to test the error bounds is to compute the FFT followed by the inverse transform, $\hat{x} = fl(F_n^{-1}fl(F_n x))$, and to evaluate $e_n = \|x - \hat{x}\|_2/\|x\|_2$. Our analysis gives the bound $e_n \leq 2n^{1/2} \log_2 n\eta + O(\eta^2)$. Figure 23.1 plots e_n and the approximate error bound $n^{1/2} \log_2 nu$ for $n = 2^k$, $k = 0:16$, with random x from the normal $N(0,1)$ distribution (the FFTs were computed using MATLAB’s `fft` and `ifft` functions). The errors grow at roughly the same rate as the bound and are on average about a factor of 10 smaller than the bound.

23.2. Circulant Linear Systems

A circulant matrix (or circulant, for short) is a special Toeplitz matrix in which the diagonals wrap around:

$$C = \begin{bmatrix} c_1 & c_n & \dots & c_2 \\ c_2 & c_1 & \dots & \vdots \\ \vdots & \ddots & \ddots & c_n \\ c_n & \dots & c_2 & c_1 \end{bmatrix} \in \mathbb{C}^{n \times n}.$$

Circulant matrices have the important property that they are diagonalized by the DFT matrix F_n :

$$F_n C F_n^{-1} = D = \text{diag}(d_i).$$

Moreover, the eigenvalues are given by $d = F_n c$. Hence a linear system $Cx = b$ can be solved in $O(n \log n)$ operations with the aid of the FFT as follows:

- (1) $d = F_n c$,
- (2) $g = F_n b$,
- (3) $h = D^{-1}g$,
- (4) $x = F_n^{-1}h$.

The computation involves two FFTs, a diagonal scaling, and an inverse FFT. We now analyse the effect of rounding errors. It is convenient to write the result of Theorem 23.2 in the equivalent form (from (23.3))

$$\hat{y} = (F_n + \Delta F_n)x, \quad \|\Delta F_n\|_2 \leq n \frac{t\eta}{1 - t\eta} =: f(n, u). \quad (23.4)$$

Steps (1) and (2) yield

$$\begin{aligned} \hat{d} &= (F_n + \Delta_1)c, & \|\Delta_1\|_2 &\leq f(n, u), \\ \hat{g} &= (F_n + \Delta_2)b, & \|\Delta_2\|_2 &\leq f(n, u). \end{aligned} \quad (23.5)$$

Equation (23.5) implies that $\hat{D} = D + \Delta_0$, $\|\Delta_0\|_2 \leq f(n, u)\|C\|_2$.

For steps (3) and (4) we have, using Lemma 3.5,

$$\begin{aligned} \hat{h} &= (\hat{D} + \Delta D)^{-1}\hat{g}, & |\Delta D| &\leq \sqrt{2}\gamma_4|\hat{D}|, \\ \hat{x} &= (F_n^{-1} + \Delta_3)\hat{h}, & \|\Delta_3\|_2 &\leq n^{-1}f(n, u). \end{aligned}$$

Putting these equations together we have

$$\hat{x} = (F_n^{-1} + \Delta_3)(\hat{D} + \Delta D)^{-1}(F_n + \Delta_2)b, \quad (23.6)$$

or, working to first order,

$$\begin{aligned} b &= (F_n + \Delta_2)^{-1}(\hat{D} + \Delta D)(F_n^{-1} + \Delta_3)^{-1}\hat{x} \\ &\approx (F_n^{-1} - F_n^{-1}\Delta_2F_n^{-1})(\hat{D} + \Delta D)(F_n - F_n\Delta_3F_n)\hat{x} \\ &\approx (F_n^{-1}\hat{D}F_n + F_n^{-1}\Delta DF_n - F_n^{-1}\Delta_2F_n^{-1}\hat{D}F_n - F_n^{-1}\hat{D}F_n\Delta_3F_n)\hat{x} \\ &=: (C + \Delta C)\hat{x}. \end{aligned}$$

We obtain a backward stability result.

Theorem 23.3. *Let $C \in \mathbb{C}^{n \times n}$ be a circulant and suppose the system $Cx = b$ is solved by the FFT process described above, where the FFT satisfies (23.4). Then the computed solution $\hat{x}$ satisfies $(C + \Delta C)\hat{x} = b$, where*

$$\|\Delta C\|_2 \leq ((1 + 2n^{-1/2})f(n, u) + 4\sqrt{2}u)\|C\|_2 + O(u^2). \quad \square$$

The conclusion is that the FFT method for solving a circulant system is normwise backward stable provided that the FFT itself is computed stably.

Although we have shown that $\hat{x}$ solves a slightly perturbed system, the perturbed matrix $C + \Delta C$ is not a circulant. In fact, $\hat{x}$ does not, in general, solve a nearby circulant system, as can be verified experimentally by computing the "(circulant backward error" using techniques from [527, 1992]. The basic reason for the lack of this stronger form of stability is that there are too few parameters in the matrix onto which to throw the backward error.

A forward error bound can be obtained by applying standard perturbation theory to Theorem 23.3: the forward error is bounded by a multiple of $\kappa_2(C)u$. That the forward error can be as large as $\kappa_2(C)u$ is clear from the analysis above, because (23.5) shows that the computed eigenvalue $\min_i |\hat{d}_i|$ is contaminated by an error of order $u \max_i |\hat{d}_i|$.

23.3. Notes and References

For a unified treatment of the many different versions of the FFT, including implementation details, see Van Loan [1044, 1992].

For a comprehensive survey of the discrete Fourier transform see Briggs and Henson [148, 1995].

The Cooley–Tukey radix 2 FFT was presented in [240, 1965], which is one of the most cited mathematics papers of all time [554, 1993, p. 171].

The history of the FFT is discussed by Cooley [238, 1990], [239, 1994] and Cooley and Tukey [241, 1993]. Cooley [238, 1990] states that the earliest known reference to the FFT is an obscure 1866 paper of Gauss in neoclassic Latin, and he recommends that researchers not publish papers in neoclassic Latin!

Theorem 23.2 is not new, but the proof is more concise than most in the literature. Early references that derive error bounds using the matrix factorization formulation of the FFT are Gentleman and Sande [434, 1966] and Ramos [859, 1971]. A full list of references for error analysis of the FFT is given by Van Loan [1044, 1992, §1.4].

Linzer [708, 1992] shows that the FFT-based circulant solver is forward stable and poses the question of whether or not the solver is backward stable. Our backward error analysis answers this question positively and therefore also proves forward stability.